

ATVED GridLock: Final Project Report

Executive Summary

ATVED (Automated Traffic Violation Enforcement Dashboard) is a comprehensive, AI-driven Smart Traffic Management Ecosystem. Its primary objective is to autonomously detect, process, and penalize traffic violations in real-time utilizing edge-deployed Computer Vision models synced to a robust cloud backend.

1. The Current Problem Statement

1.1 No Automation in Traffic Violation Detection

Currently, traffic enforcement relies heavily on manual monitoring by traffic police officers. This manual process is inefficient, prone to human error, and unable to scale to monitor multiple intersections 24/7. Many violators escape undetected simply because an officer cannot observe every vehicle simultaneously.

1.2 Manual Fine Collection Leads to Corruption

When a traffic violation is caught manually, the fine is often collected in cash or settled "on the spot." This lack of a digitized, transparent audit trail creates a massive loophole for bribery and corruption. The state loses revenue, and the deterrent effect of traffic laws is severely compromised.

2. Proposed Model

The ATVED Solution

To address these issues, we propose the **ATVED GridLock System**, an end-to-end automated platform that replaces manual observation and cash fines with AI vision and digital enforcement.

- **AI-Powered Edge Cameras:** Video feeds from traffic cameras are processed in real-time using advanced deep learning models (YOLOv8) to autonomously detect violations like speeding, riding without a helmet, and triple-riding.
 - **Automated Digital Penalties:** Once a violation is confirmed, the system uses a Hybrid OCR pipeline to extract the license plate, matches it to the vehicle database, and automatically deducts the fine from the driver's linked bank account while issuing a digital E-Challan.
 - **Zero Human Intervention:** By removing the human element from both the detection and the fine collection process, corruption is eliminated, and enforcement scales to 100% of the camera's field of view.
-

3. The Problem It Will Solve

- **Eradication of Bribery:** With automated, direct-to-bank fine deductions and digital E-Challans, on-the-spot cash bribery is eliminated.
 - **24/7 Enforcement:** AI never sleeps. The system provides continuous, unbiased enforcement at all monitored intersections.
 - **Behavioral Deterrent:** By introducing a "Traffic Score" that lowers with each violation and multiplies future fines, the system acts as a severe deterrent for repeat offenders.
 - **Resource Optimization:** Traffic police are freed from manual observation duties and can be redeployed to handle severe incidents, accidents, or complex traffic management tasks.
-

4. Complete Architecture & Implementation

The ATVED system is divided into two primary loops: **The Edge AI Loop** and **The Cloud Processing Loop**.

4.1 Edge AI Loop (Computer Vision)

1. **Frame Ingestion:** Live video streams (RTSP/MP4) are ingested frame-by-frame.
2. **Tracking & Detection:**
 - YOLOv8 detects vehicles (cars, motorcycles, buses) and cross-references them with nested classes (e.g., helmets, seatbelts, persons).
 - ByteTrack assigns persistent tracking IDs to ensure vehicles are uniquely tracked across frames without duplicate counting.
3. **Violation Analysis:** Spatial and temporal logic runs on tracked objects:
 - *Speeding:* Calculated via pixel displacement over time.
 - *Triple Riding:* Triggered if 3 or more person bounding boxes overlap a single motorcycle.
 - *No Helmet:* Triggered if a motorcycle rider is classified without a helmet.
4. **Hybrid OCR Pipeline (v2):**
 - The license plate is cropped, given a 15% black canvas padding, and upscaled using cubic interpolation.
 - Contrast is enhanced via CLAHE (LAB color space) and **Adaptive Gaussian Thresholding** (blockSize=21, C=8) is applied to perfectly preserve character loops (e.g., 3, 6, 8, 9).
 - `fast-plate-ocr` reads the text.
 - *Strict Regex Fallback Gate:* The output is evaluated against strict Indian license plate regex patterns. If it structurally fails validation, EasyOCR is dynamically triggered on the un-binarized crop to recover lost data.
5. **Payload Dispatch:** The AI constructs a JSON payload (`camera_external_id`, `violation_type`, `plate_text`, `confidence_score`) and posts it to the backend API.

4.2 Cloud Processing Loop (Backend API)

1. **Ingestion & Driver Lookup:** The FastAPI server receives the payload and matches the plate text to a registered `Driver`.

2. **Dynamic Fine Mitigation (Good Samaritan Loop):** The system checks the driver's `eligible_for_mitigation` flag. If `True`, a 10% discount is applied, and the flag is reset.
 3. **Score Deterrent System:** Points are deducted from the driver's score based on a `PENALTY_MATRIX`. Lower scores dynamically multiply the base fine of future violations (e.g., `< 200` score = $3.0 \times$ fine multiplier).
 4. **Financial Deduction:** The fine is automatically deducted from the driver's simulated `bank_balance`.
 5. **Automated Dispatch:**
 - If the AI Confidence Score is $\geq 85\%$, the system synchronously generates an E-Challan PDF and dispatches an SMS with the link.
 - If $< 85\%$, it is logged silently for human review.
-

5. Pros

- **Complete Transparency:** Every violation is accompanied by a timestamped, annotated photographic evidence crop.
 - **High Accuracy & Fault Tolerance:** The multi-tiered Hybrid OCR pipeline and temporal voting mechanics ensure that misreads are extremely rare.
 - **Dynamic Deterrence:** The Traffic Score system inherently punishes repeat offenders exponentially harder than first-time violators, encouraging long-term behavioral change.
 - **Scalability:** The asynchronous FastAPI backend can handle thousands of simultaneous violation payloads from hundreds of cameras.
-

6. Current Limitations

1. **Severe Weather Dependency:** Extremely heavy rain, thick fog, or direct sun glare directly into the camera lens will degrade OCR accuracy and YOLO object detection confidence.
 2. **Hardware Requirements:** The hybrid OCR pipeline (specifically the EasyOCR fallback and running multiple YOLO models simultaneously) requires substantial GPU VRAM. Running multiple high-res camera streams on a single node without a dedicated AI accelerator (like an NVIDIA T4 or A100) causes frame drops.
 3. **Non-Standard Plates:** Custom fonts, broken plates, heavily mud-covered plates, or non-reflective plates pose a significant challenge to the OCR, resulting in lower confidence scores that require manual review.
-

7. Future Improvements

- **Edge-Cloud Split Computing:** Migrate the heavy YOLO inference to dedicated TensorRT edge devices (like NVIDIA Jetson Orin) mounted directly on the traffic poles, transmitting only cropped evidence images to the cloud for OCR and billing.
- **Night-Vision & IR Integration:** Train models specifically on Infrared (IR) camera feeds to maintain 100% detection accuracy during complete darkness without relying on streetlights.
- **Blockchain Audit Trail:** Log every generated E-Challan hash onto a lightweight

blockchain ledger to cryptographically prove that a violation record was never tampered with or deleted by a corrupt official after generation.

- **Automated Video Evidence:** Instead of a single static image, attach a 3-second looping GIF/MP4 of the violation to the E-Challan to provide irrefutable proof (crucial for dynamic violations like speeding or red-light running).